

IA: MACHINE LEARNING FOUNDATIONS

SESSION 7 – Classification

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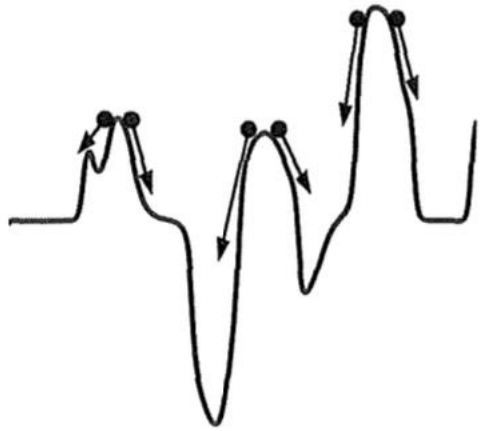
AGENDA

1. Gradient Descent
2. Assignment



1. GRADIENT DESCENT

- Gradient Descent is a general technique for minimizing an error function.
- It can be used to optimize many machine learning models, and it has particularly nice properties for regression and classification.
- **It works by adjusting parameters in the direction of the negative gradient.**
- The error function is a “surface” in a high-dimensional space.
- Depending on where you start you may end up at a *local minimum*.



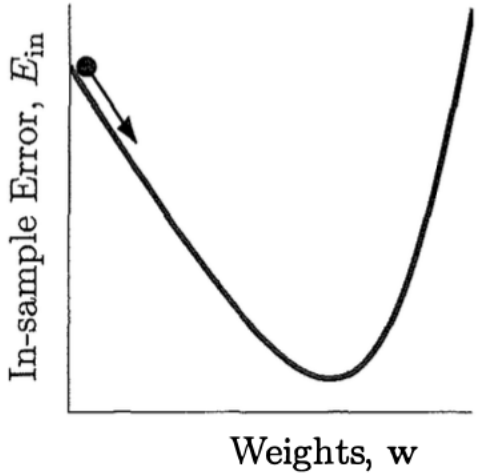
How Gradient Descent Works

- Calculates the gradient of the error function with respect to the parameters.
- Updates the parameters in the opposite direction to the gradient.
- The step size is determined by the learning rate.
- The process is repeated until a minimum or stopping criterion is reached.

1. GRADIENT DESCENT

EXAMPLE

- Logistic Regression & Cross-Entropy Error
- We won't have local minima and we just must be sure we stop at the bottom of the valley.
- How do we "roll" down the surface?



$$\boldsymbol{w}(t+1) = \boldsymbol{w}(t) + \eta\hat{\boldsymbol{v}} \qquad \hat{\mathbf{v}} = -\frac{\nabla E_{\text{in}}(\mathbf{w}(0))}{\|\nabla E_{\text{in}}(\mathbf{w}(0))\|} \qquad E_{\text{in}}(\mathbf{w}) = \frac{1}{N}\sum_{n=1}^{N} \ln\left(1 + e^{-y_n \mathbf{w}^{\mathsf{T}} \mathbf{x}_n}\right)$$

Logistic regression algorithm:

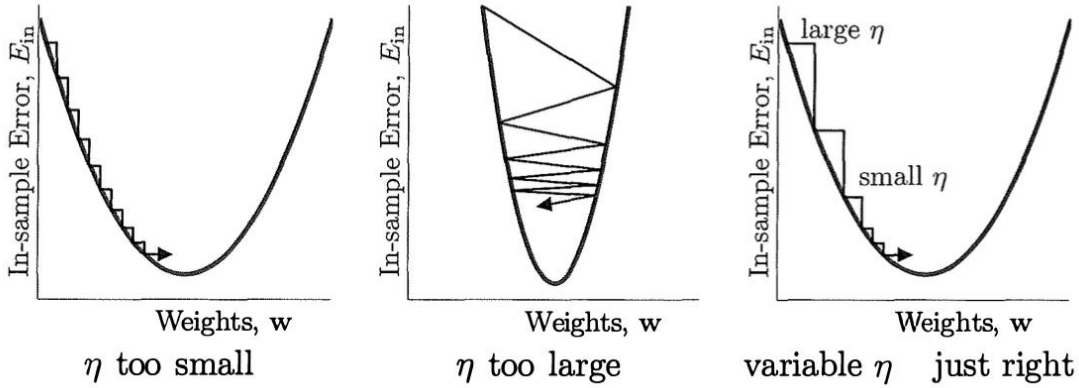
1: Initialize the weights at time step $t = 0$ to $\mathbf{w}(0)$.
2: **for** $t = 0, 1, 2, \ldots$ **do**
3: Compute the gradient

$$\mathbf{g}_t = -\frac{1}{N}\sum_{n=1}^{N} \frac{y_n \mathbf{x}_n}{1 + e^{y_n \mathbf{w}^{\mathsf{T}}(t)\mathbf{x}_n}}.$$

4: Set the direction to move, $\mathbf{v}_t = -\mathbf{g}_t$.
5: Update the weights: $\mathbf{w}(t+1) = \mathbf{w}(t) + \eta\mathbf{v}_t$.
6: Iterate to the next step until it is time to stop.
7: Return the final weights $\mathbf{w}$.

1. GRADIENT DESCENT

- **How large a step should one take at each iteration?**



- **How do we initialize $w(0)$?**
  - Zeros
  - Randomly
  - Each weight independently from a Normal distribution with zero mean and small variance

- **How do we decide when to stop?**
  - Upper bound on number of iterations
  - When $\boldsymbol{g_t}$ drops below a threshold
  - When the error is small, and the change is small
  - A combination

1. GRADIENT DESCENT

- **Gradient Descent Variants**
  - Batch Gradient Descent uses the entire data set per iteration.
  - Stochastic Gradient Descent updates with each individual sample.
  - Mini-batch Gradient Descent combines the advantages of both methods.
  - Variants impact the speed and stability of learning.

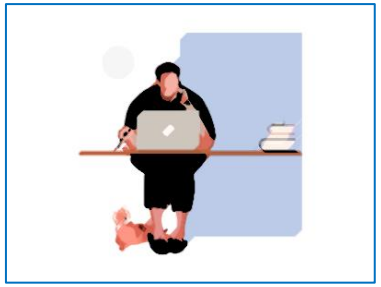
- **Applications**
  - It trains models such as linear and logistic regression.
  - It optimizes deep neural networks.
  - It is used in clustering and dimensionality reduction techniques.
  - It is the basis for many modern machine learning algorithms.

AGENDA

2. Gradient Descent
3. Assignment



3. ASSIGNMENT



PROBLEM

Use the “session_7_dataset.xlsx” for this assignment. You will find data from several properties in Madrid as well as some historical and sociodemographic data.

1. Analyze this dataset to estimate property prices as a regression problem.
    - Use algorithms seen so far in class.
2. Analyze data and redefine the problem to create a classification one.
    - Use the following algorithms and analyze results:
        - Perceptron Learning Algorithm
        - Logistic Regression
        - Generative Models (LDA, QDA)
        - KNN
3. Apply other methods you may know and compare results with the previous ones.


THANK YOU